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GIA - IAA

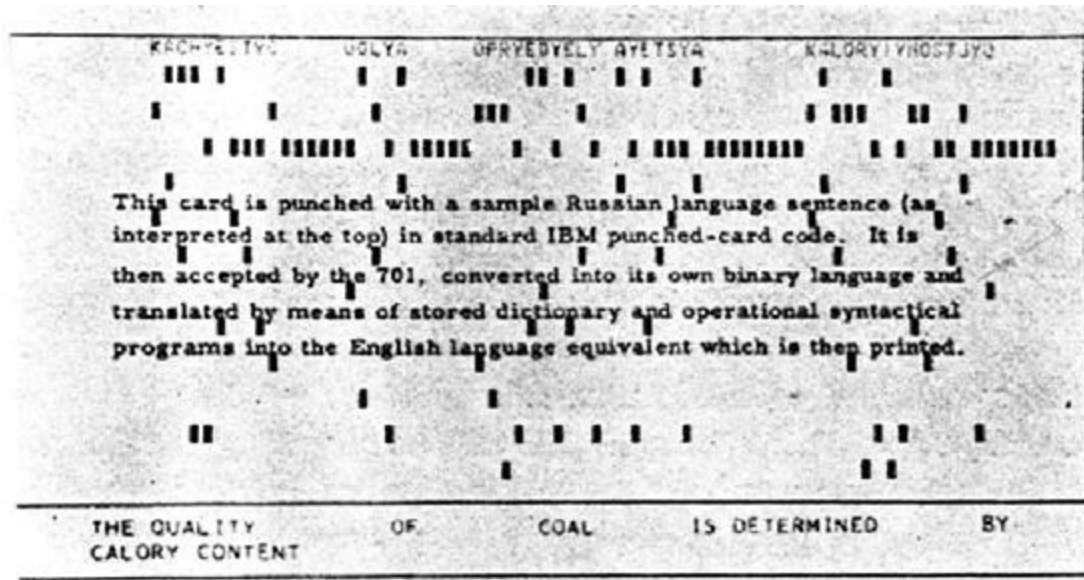
Introduction to Machine Learning

2. Basic concepts

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The Winters of AI

- ❖ The hype was killed before
 - XOR!
 - Hard-coding experts!
- ❖ This time AI has, and will, deliver
- ❖ ~~Winter rationalization~~



The Golden Age of ML

- ❖ Monetize data & compute
- ❖ Flashy & noisy
- ❖ Emergence of representation learning
- ❖ Not a lab game anymore



Current Market

- ❖ OpenAI (MSN), DeepMind (Google), Midjourney
- ❖ Meta, HuggingFace, stability.ai
- ❖ EleutherAI, LAION, mozilla
- ❖ Stanford, DAIR, CohereAI, ...



Limits of ML

- ❖ Rely on data and digitalization
 - We can rarely choose
- ❖ Can't learn what isn't there / Can't not learn what is there
 - Bias party
- ❖ Algorithmic short-cuts & human complexity



This class is not

- ❖ This world
- ❖ This city (x2)
- ❖ This generation (x2)
- ❖ This campus (x2)

Be like the data, not the data

❖ Model

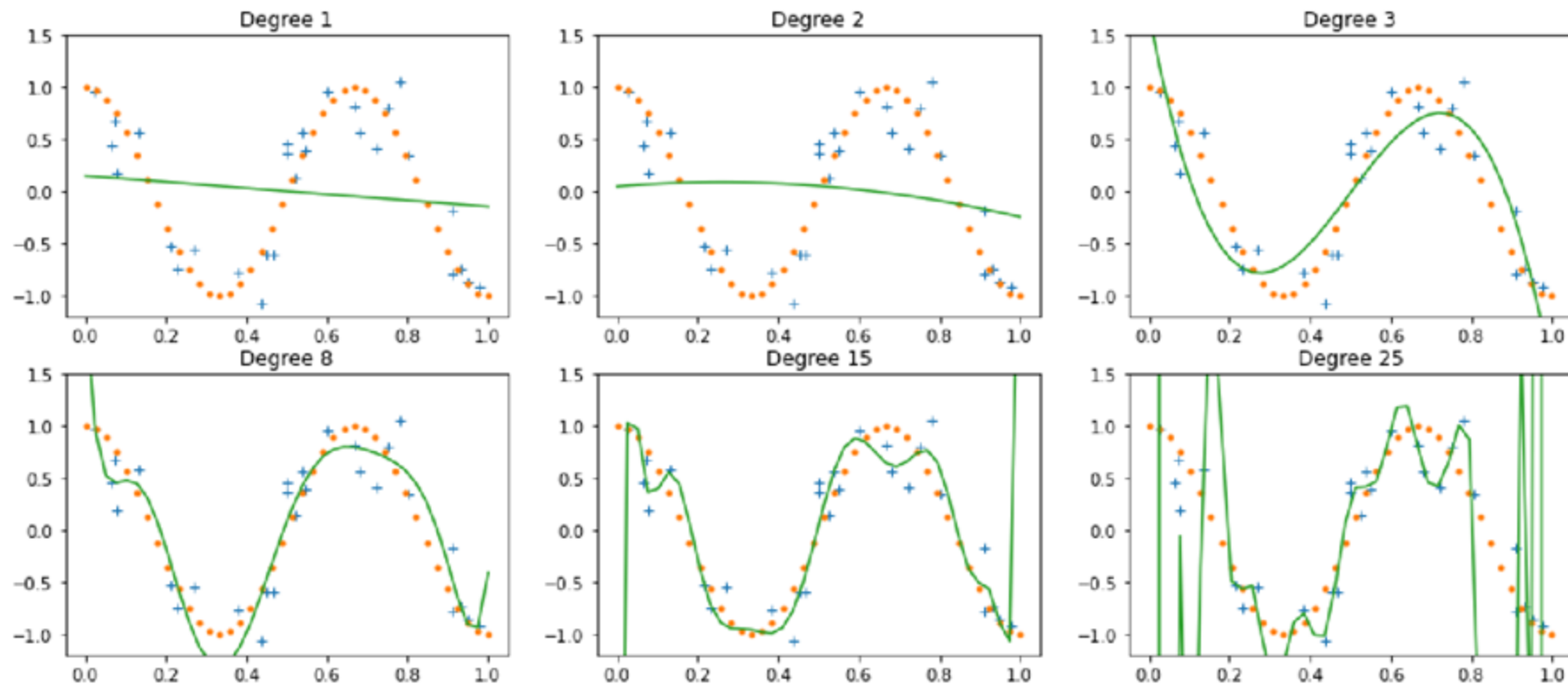
- Complexity
- Data exposition



❖ Model

- Bias (simplicity & underfit)
- Variance (complexity & overfit)

Under, Fit, Over



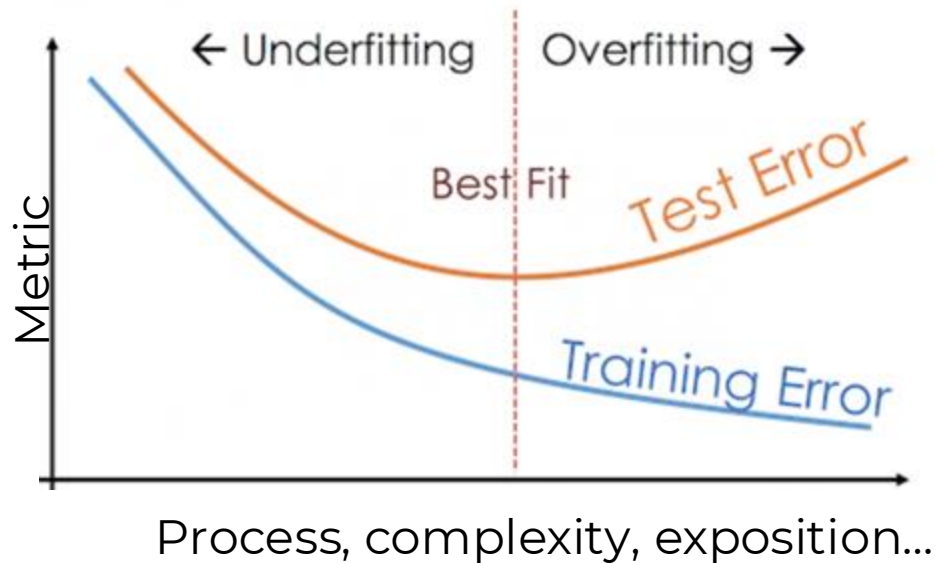
Generalization & Overfitting

- ❖ Learning *right/wrong*
 - Bad luck doesn't favour the lazy
 - Sample matters. Order matters.
 - Non-conforming stuff
- ❖ AI big stopper?
- ❖ The eternal elephant in the room



do this RIGHT

- ❖ Train (model)
- ❖ Val (you)
- ❖ Test (3rd person)
- ❖ Don't fool yourself



Mental Tattoo

“All models are overfitted, some are usefully overfitted”

The many dangers

- ❖ Train / Val / Test / Var correlations
- ❖ Data/Model complexity
- ❖ Data order
- ❖ The seed
- ❖ Suicide by non-reproducibility
- ❖ Trust no Train
- ❖ Bias? Y/N How big?
- ❖ Imbalance



Mental Tattoo

“Correlation does not imply causation”



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Let's play a game

Data, where to start/end

- ❖ Goal: Predict the score of a student
 - Which variables?
 - As a supervised problem
 - Classification
 - Regression
 - As an unsupervised problem
 - Feature analysis & User profiling



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Incomplete guidelines

Survival pre-processing

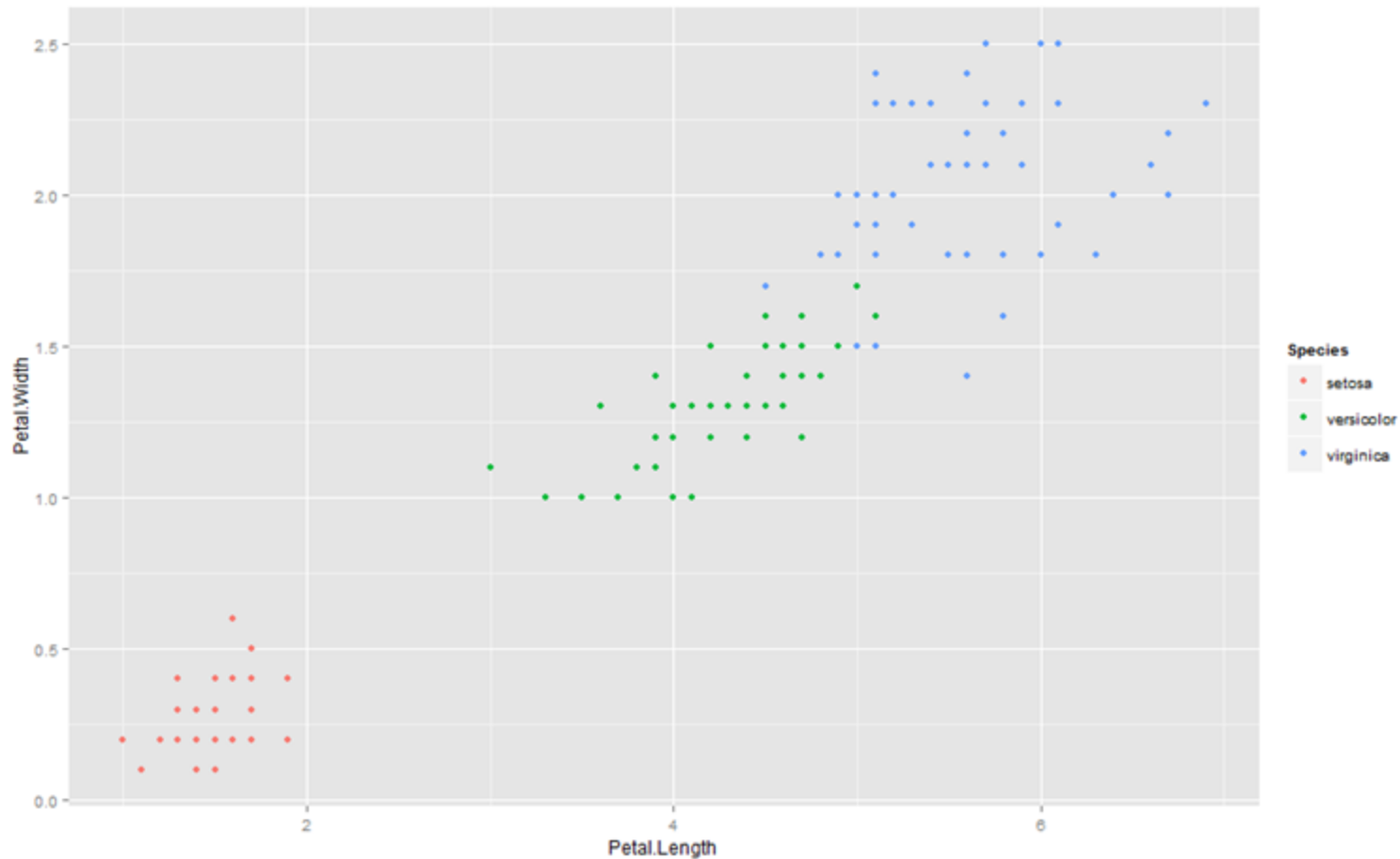
❖ Full pre-processing-> ME/SM

❖ Never enough distributions

- Who's missing?
- Who's weird?



Always look at the data

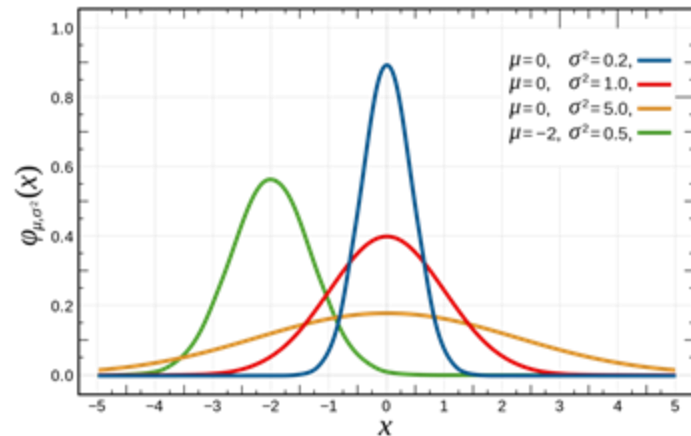
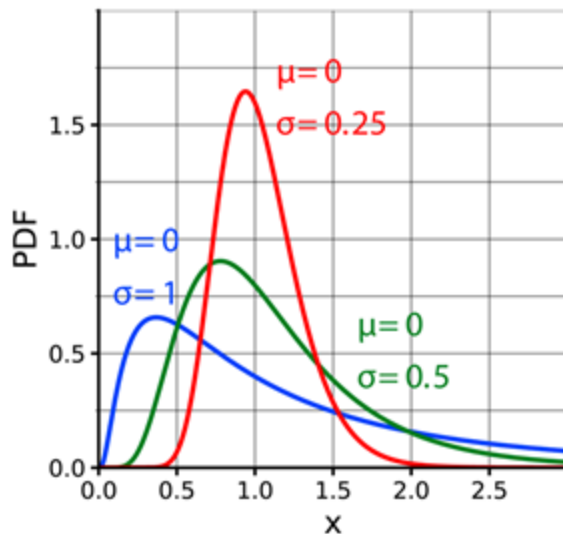


Ain't normal being Normal

- ❖ Gaussian
 - Mean 0, Std 1

- ❖ Log-Normal

- ❖ Power-law



- ❖ Means & deviations

Assumptions

- ❖ I.I.D. Samples
 - Independent
 - Order & Source bias
 - Identically distributed
 - Rough, yes. Mutating, no
- ❖ Prob./likelihood/some ML must



Data Normalization

❖ Mix-Max scaling

- range, rare (e.g., all in [0,1])

$$x_{new} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

❖ Z-score standardization

- Gaussianizing a feature
- Mean 0, Std. Dev 1

$$z = \frac{x_i - \mu}{\sigma}$$

Transformations

Features

- ❖ Discretization
- ❖ Binarization
- ❖ One-hot encoding
- ❖ Label encoding

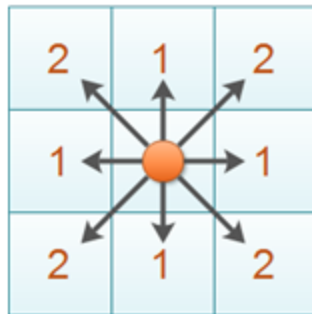
Samples

- ❖ Data Augmentation
- ❖ Cropping
- ❖ Sliding window
- ❖ ...

Distances

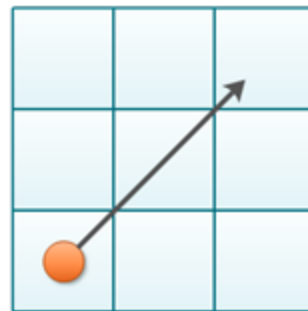
- ❖ L-0 (outlier resistant)
- ❖ L-1 (Manhattan)
- ❖ L-2 (Euclidean)
- ❖ L-inf (outlier only)

Manhattan Distance

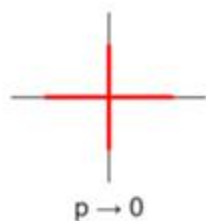


$$|x_1 - x_2| + |y_1 - y_2|$$

Euclidean Distance



$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$



$$\|x\|_0 = (\sum_i |x_i|^0)^0$$

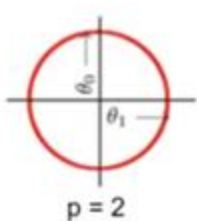
The number of non-zero parameters



$$\|x\|_{0.5} = (\sum_i |x_i|^{0.5})^2$$



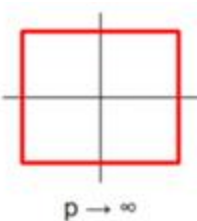
$$\|x\|_1 = (\sum_i |x_i|^1)^1$$



$$\|x\|_2 = (\sum_i |x_i|^2)^{1/2}$$



$$\|x\|_4 = (\sum_i |x_i|^4)^{1/4}$$



$$\|x\|_\infty = (\sum_i |x_i|^\infty)^0$$

The size of the largest parameter

Comparing distributions

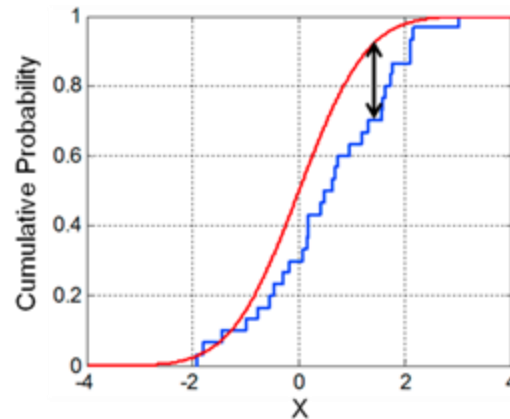
- ❖ Kolmogorov-Smirnov

- Max gap (CDF, !multimodal)

- ❖ Kullback-Leibler

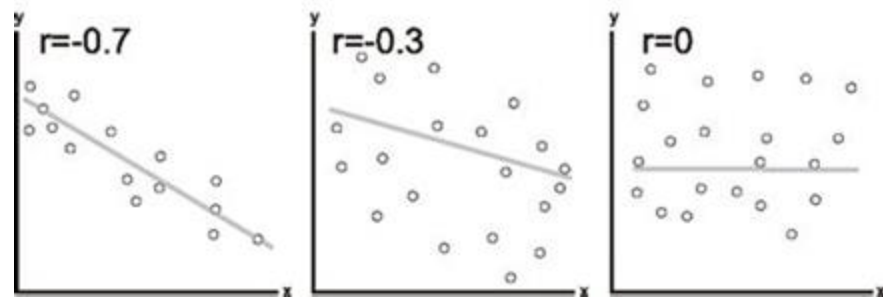
- Likelihood comparison (entropy, !sym, !tri)

- ❖ And many others...



Correlations

- ❖ Pearson (distance)
 - Cont. wo/ outliers
 - Assumes linearity & Gaussianity



- ❖ Spearman (order)
 - Cont w/ outl., Ordinal
 - ~~Linear. Normal~~

❖ ...

The Scientific Method in ML

- ❖ Diagnose at the model
 - Plots, mats, top of the line & XAI
- ❖ Make a hypothesis
- ❖ One test at a time
- ❖ Repeat



Top-of-the-line metrics

- ❖ Never settle for one
- ❖ Check the peers
- ❖ Ask the guide

		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Total population = P + N		
	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection

Curves and Matrices

$$\frac{TP}{TP + FP}$$

How many of my predictions as T are actually T?

❖ Precision

$$\frac{TP}{TP + FN}$$

How many T am I predicting as T?

❖ Recall

$$\frac{TP + TN}{P + N}$$

How many predictions are right?

❖ Accuracy

$$2 \times (\text{Precision} \times \text{Recall})$$

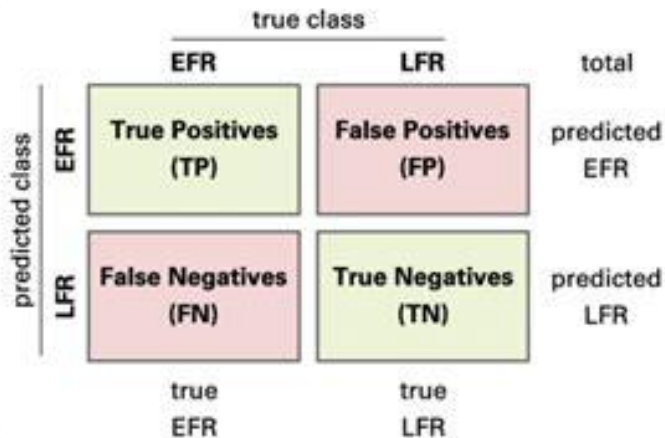
Trade-off precision and recall

❖ F1-score

$$\text{Precision} + \text{Recall}$$

Confusion Matrix

- ❖ Make one and compute metrics



Curves

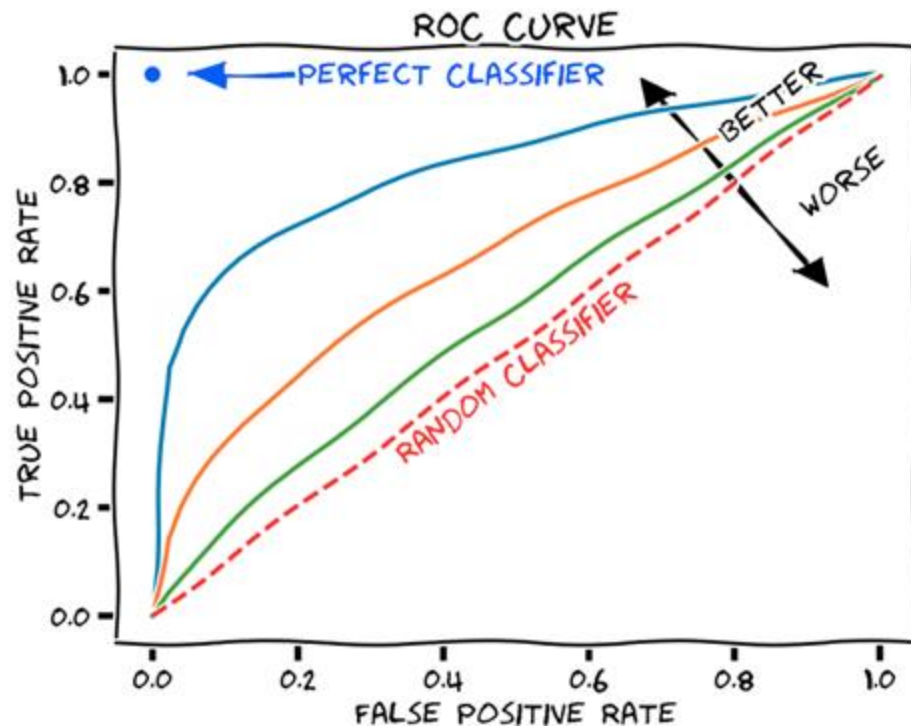
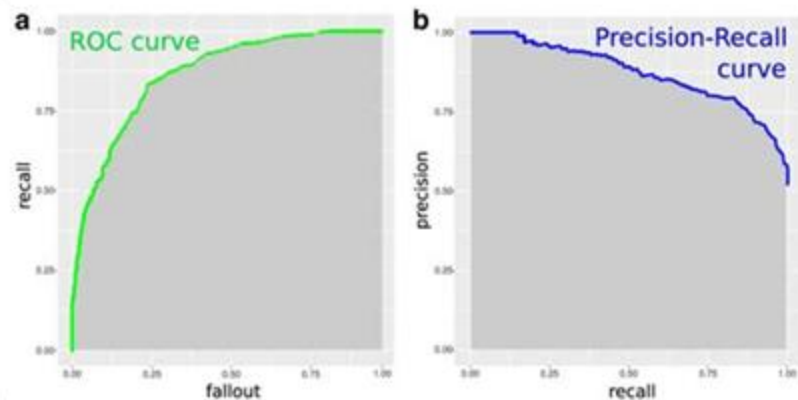
False Positive Rate = FP/N

True Positive Rate = TP/P

❖ ROC-AUC

- Misleading in imbalance

❖ PR-AUC



Imbalanced datasets are the worst

- ❖ Uneven data
 - Sampling bias
- ❖ High impact on model learning
 - Learn the mode
- ❖ High impact on model evaluation
 - Assess the mode
 - Mitigations: Average per class metrics, conf. matrices, plots

For 99% of real data

❖ Undersampling

- Losing signal

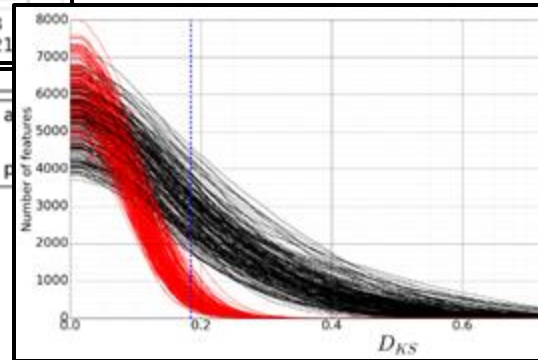
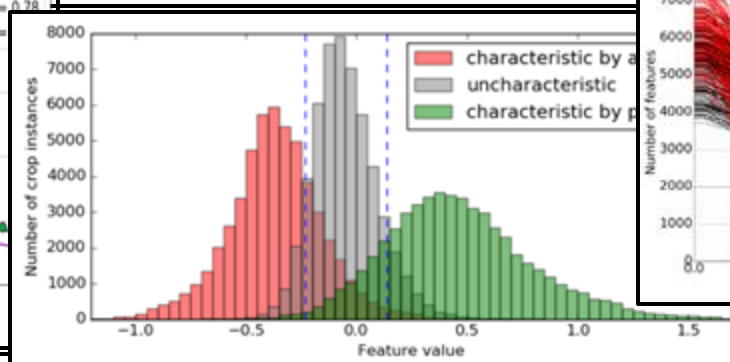
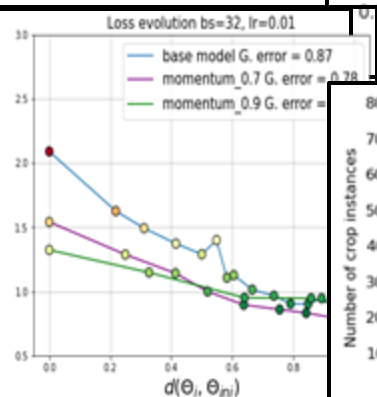
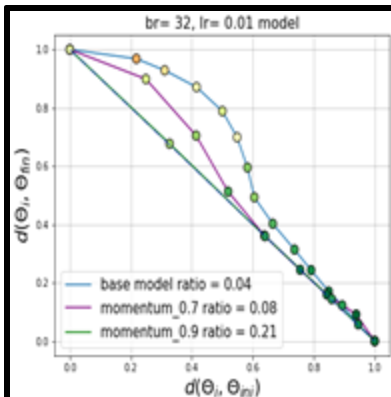
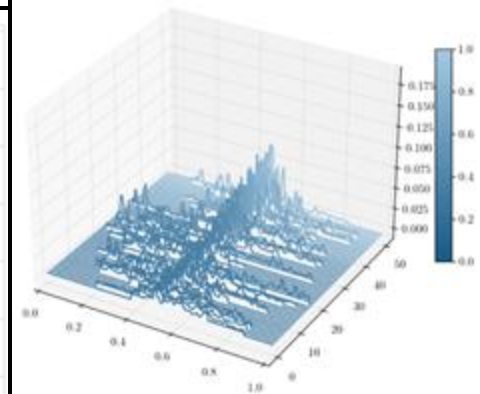
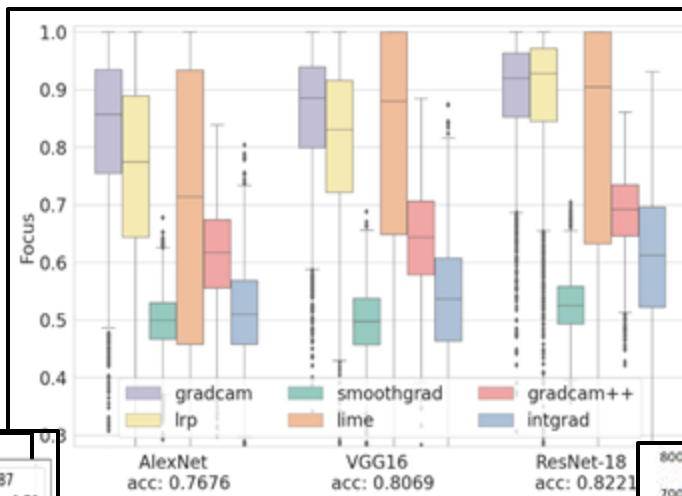
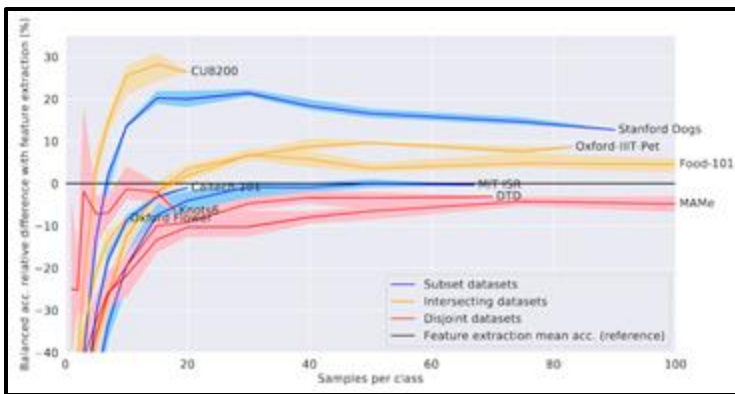
❖ Class weights (model-wise)

❖ Oversampling

- Adding redundancy & noise (OF)
- SMOTE

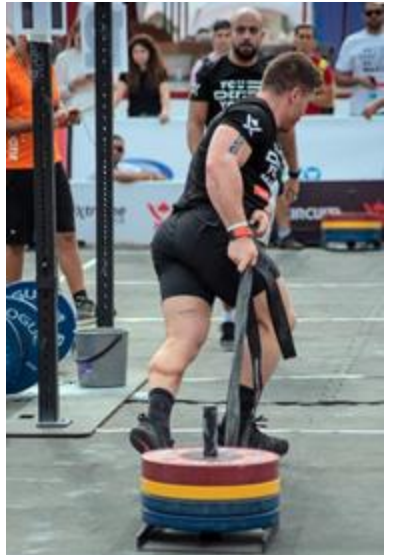
KNN + combination in feature space + random middle point

Plots: Axis, legends & captions



Ablation studies

- ❖ Pulls vs dead weight
- ❖ Occam rulez
- ❖ Mind the footprint



Evaluating through I/XAI

- ❖ A magic wand can't be trusted
- ❖ For things that really matter, start with this
- ❖ For lesser things, it can go second
- ❖ A top-of-the-line AND an I/XAI



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Let's play a game

ML tasks

- ❖ Goal: Predict the score of a student
 - Variable types, ranges, missings, outliers, distribution
 - Binary, categorical, continuous, ordinal, ...
- ❖ This is all personal data



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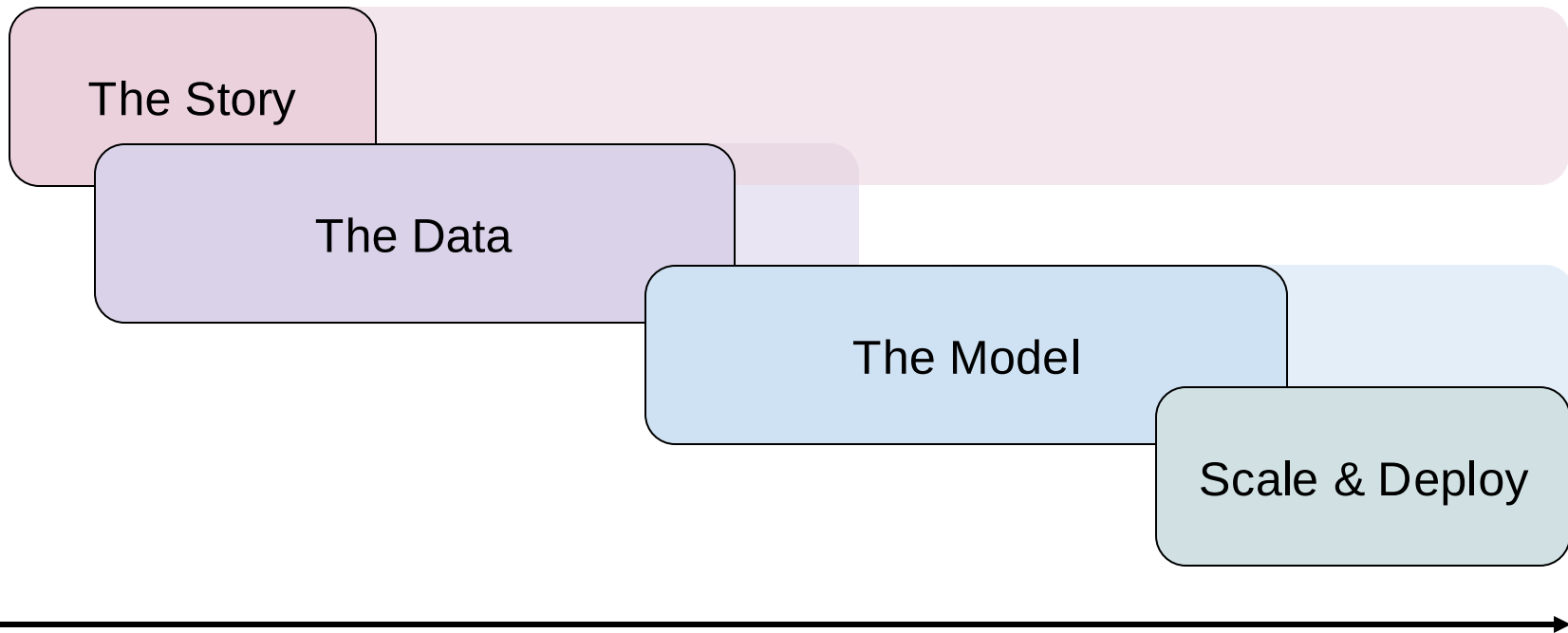
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Machine Learning steps

this is far from over

The ML Cycle



The Story

- ❖ What & why
 - The moon
 - All of it
 - Curiosity & Imagination



The Data

❖ The Story always in mind

- Types
- Volume
- Gathering & Origin
- Danger

and the ugly truth

The Model

- ❖ Being an engineer
- ❖ Baseline first
- ❖ Back to data
- ❖ Bring a local guide along
- ❖ Never worry enough
- ❖ Model selection & ensembling



Scaling & Deploying

- ❖ User's rule: Performance & Understanding
- ❖ New shoes hurt
- ❖ Containers & Resources